

ComplexGitSync 0002.21– User Guide

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Contents

1	User Guide	2
2	CLI Overview	2
3	Lifecycle Contract	2
4	Commands	2
4.1	initialise	3
4.2	bootstrap	3
4.3	discover	4
4.4	configure	5
4.5	create-cgs	5
4.6	clean-init	5
4.7	purge	5
4.8	pull	5
4.9	pull-force	6
4.10	clone	6
4.11	checkout	6
4.12	add	6
4.13	commit	7
4.14	push	7
4.15	freeze-release	7
4.16	freeze	7
4.17	import-submodules	8
4.18	verify	8
4.19	launch-release	9
4.20	view-tree	9
4.21	status	9
5	Document Formats	9
5.1	Local Authoring Spec (.cgs)	10
5.2	Repository Placement and Nested Discovery	11
5.3	Git Tree State Snapshot (.gts)	12
5.4	Local Git Register and Sync Ledger (.lgr)	13
6	Worked Examples: From CGSil1 to cawaqsviz	13
6.1	Level 1 — CGSil1: the minimal synthetic topology	14
6.2	Level 2 — cawaqs: the same minimal style, at scale	14
6.3	Level 3 — cawaqsviz: three ways to the same .cgs	15
6.3.1	3a — Write it by hand	15

6.3.2	3b — Derive it with <code>discover</code> (autodiscover mode)	15
6.3.3	3c — Migrate it with <code>import-submodules</code>	16
6.3.4	A shared caveat across all three	16
7	Configuration and Environment	16
8	Error Reference	17
9	Troubleshooting	17

1 User Guide

This chapter is the authoritative reference for every user-facing command, configuration option, document format, and error condition exposed by `ComplexGitSync`.

2 CLI Overview

The single entry-point is the `cgitsync` command.

cgitsync is a Pixi task, not a globally installed executable. This project is developed and run exclusively through Pixi (`pixi install` once, per checkout) — there is no `pip install` that puts a bare `cgitsync` on `$PATH`. Every invocation shown in this chapter as `pixi run cgitsync ...` must be typed exactly that way; a bare `cgitsync ...` will not be found by the shell.

```
$ pixi run cgitsync <command> [options]
```

Running `cgitsync` without a command prints a help summary and exits with code 0.

3 Lifecycle Contract

This guide follows the simplified canonical lifecycle:

1. `initialise(.cgs)` → clone all repos → `READY` (*new project*)
`initialise(.gts)` → restore from snapshot → `READY` (*existing project*)
2. `pull(.cgs|.gts)` → resync an existing tree
3. `checkout(.gts)` / `add` / `commit` / `push` / `tag` → git operations on the tree
4. `freeze` → emit the next `.gts` id and update `<Project_name>.lgr`

The project-local `.lgr` register and sync ledger are both active (T24, T32). Every synchronisation operation is automatically recorded as an immutable DAG event in the `[[ledger]]` section of the `.lgr` file.

4 Commands

The CLI exposes three command groups once a `GitTree` is `READY`. The minimalist level is `initialise`, `bootstrap`, `clean-init`, `freeze-release`, `freeze-release-force`, `status`, `view-tree`, and `launch-release`. The expert level exposes the individual operations: `branch`, `checkout`, `purge`, `validate`, `clone`, `pull`, `pull-force`, `add`, `commit`, `push`, `tag`, `freeze`, `import-submodules`, and `verify`. The configuration level, `discover`, `configure` and `create-cgs`, authors a `.cgs` specification without requiring existing state. Redundant inspection aliases such as `print`, `view-operation`, and `validate-topology` are not part of the public CLI surface.

4.1 initialise

```
$ pixi run cgitssync initialise <spec.cgs> [--output-path DIR] [--commit-  
  gitignore]  
  [--force-gitignore-sync] [--git-user-name NAME] [--git-user-email  
    EMAIL]  
  [--force-protocol {ssh,https}]  
$ pixi run cgitssync initialise <snapshot.gts>  
$ pixi run cgitssync initialise --project NAME --repo PROVIDER:OWNER/REPO  
  [--repo ...] [--output-path DIR]
```

Unified initialisation entry point. Dispatches on file extension:

- **.cgs** source (clone mode): clones the full repository tree. The optional `-output-path` controls `CGSPATH`; `CGSHOME` is derived as `CGSPATH/<project-name>`. When omitted, `CGSPATH` defaults to `../...` On success, ends in `READY`. If clone setup fails because stale generated state is still present, the CLI prints `Try clean-init method`.
- **.gts** source (restore mode): loads a saved snapshot directly. On success, ends in the lifecycle state recorded in the snapshot.
- Direct CLI authoring: requires `-project` and at least one repeatable `-repo`. It is mutually exclusive with a source file. The CLI collects values only and calls `ComplexGitSyncClient.configure()`; that reusable Python API delegates parsing, normalization, and validation to `cgs_format.py` and produces the same canonical `CgsDocument` as an equivalent `.cgs` file.

All initialization paths end in a `READY` tree or fail explicitly.

Before a `.cgs` source is confirmed ready, every repository with children (root, or any nested repository that itself has further nested children) is safely pulled parent-first and has its `.gitignore` updated with the relative path of each immediate child — nested repositories are plain independent clones rather than gitlinks, so without this a parent's `git status/git add` would otherwise see a child's working tree as ordinary untracked content. This is logged as the `GT-GITIGNORE` phase, after `GT-CLONE`. If the safe pull for one of these repositories fails, `initialise` raises immediately and nothing is written; pass `-force-gitignore-sync` to instead fall back to a pull-force recovery for that one repository (never a force-push). By default nothing is staged, committed, or pushed by this step — it only writes the file and prints what changed. Pass `-commit-gitignore` to explicitly approve staging (`.gitignore` alone), committing (with a message listing exactly which children were added), and pushing each changed repository. The commit identity defaults to local git config; `-git-user-name/-git-user-email` override it and persist the override to `CGSHOME/.cgitsync/master.toml` via `MasterConfig`, so later invocations on the same workspace pick it up without repeating the flags.

`-force-protocol {ssh,https}` is an expert option, typically used systematically in CI rather than for everyday development. It overrides `access_protocol` in memory for every repository this run clones — including ones discovered later from a nested `.cgs` living in a different, separately-cloned repository — so no `.cgs` file is ever read differently or written to. Forcing `https` is the usual CI case: it removes the dependency on an SSH key or agent being present on the runner, a dependency that otherwise differs between a `push` run and a `pull_request` run (GitHub Actions withholds repository secrets from a pull request opened from a fork). Omitted (the default), each repository follows whatever protocol its own `.cgs` entry declares, exactly as without the flag.

4.2 bootstrap

```
$ pixi run cgitSync bootstrap <spec.cgs> <project_name> [--cgs-path DIR]
  [--force-protocol {ssh,https}]
```

Clones the full repository tree from a `.cgs` source into an isolated `CGSHOME`, for running `cgitSync` from its own standalone clone rather than nested inside the project tree it manages — one `ComplexGitSync` clone reused across many projects. It delegates to the same `ComplexGitSyncClient.clone_cgs` pipeline as `clone` below, and so shares its branch-fallback and nested-discovery behaviour, but differs from both `clone` and `initialise` in two ways:

- `project_name` is a required positional argument, not inferred from the `.cgs` document. It always forms the final path segment of `CGSHOME`, regardless of the specification’s own project name.
- `CGSPATH` does not default to `../..` relative to the current directory (that default assumes `ComplexGitSync` is cloned inside the tree it manages — the nested-mode case `initialise` is for). When `-cgs-path` is omitted, `CGSPATH` instead defaults to a fresh `$HOME/.cgs/CGS<timestamp>/` directory, created automatically if it does not exist, so a bootstrapped project can never land inside `ComplexGitSync`’s own clone.

`-force-protocol {ssh,https}` behaves exactly as described under `initialise` above, applying here to every repository the run clones — including the root, which `bootstrap` (unlike `initialise`) also clones from scratch rather than assuming already checked out.

4.3 discover

```
$ pixi run cgitSync discover [ROOT] [--write FILE] [--max-depth N]
```

Scans `ROOT` (default: the current directory) for git repositories and drafts a `.cgs` from what is checked out, through the `ComplexGitSyncClient.discover_repos()` facade. This is the entry point for adopting a project that exists on disk but has no `.cgs` describing it yet.

The command is a pure read of the filesystem and of each repository’s git config: nothing is cloned, fetched, staged, or modified, and no network call is made. Without `-write` it only prints what it found, matching the “report first, write only when asked” posture of `-commit-gitignore` and `import-submodules -apply`.

The walk descends up to `-max-depth` levels (default 5; `ROOT` itself is depth 0), treating every directory that contains a `.git` entry as a repository. It never descends *into* a `.git` directory: for a submodule the real git directory lives at `<parent>/..git/modules/<name>` while the child’s own `.git` is a file, so walking into it would report the same repository twice.

For each repository, `origin`’s URL is read and converted to the canonical `provider:owner/repository` shorthand through the existing `parse_repo_id()` grammar — this command adds no second parser. `relative_path` is taken straight from the walk rather than from a repository name, so a child mounted at `external/Thing` is recorded there and not at `Thing`. `nested_config` is left unset on every entry: the default `auto` already resolves cleanly whether or not a repository carries its own `*.cgs`.

Repositories with no `origin`, or whose remote URL does not resolve to a registered provider, are reported as warnings and left out of the draft. They are never guessed at: a draft that silently invented an address would be worse than one that states what it could not determine. Always review the generated file, then `validate` it.

Only what is *checked out at scan time* can be found. A repository cloned without `-recurse-submodules` leaves its submodule paths as empty directories, and those are correctly not reported; recovering them from git metadata instead is `import-submodules`’ job.

4.4 configure

```
$ pixi run cgitsync configure [--output FILE]
```

Interactively prompts for a project name, default branch, and one or more repositories (provider, owner, name, and per-repository overrides), then writes the resulting `.cgs` file through the same offline `ComplexGitSyncClient.configure()` facade used by `initialise -project/-repo` and `create-cgs`. The provider prompt accepts `github`, `gitlab`, `codeberg`, or `custom`; `custom` additionally prompts for the provider URL. When `-output` is omitted, the CLI prompts for a destination path, defaulting to `<project-name>.cgs`.

4.5 create-cgs

```
$ pixi run cgitsync create-cgs --project CGSil1 \  
  --repo github:flipoyo/ComplexGitSync \  
  --repo codeberg:GX4G/GX4G \  
  --output CGSil1.cgs
```

Builds the canonical `CgsDocument` through the offline format pipeline and serializes concise `.cgs` TOML. It performs no Git or network work.

4.6 clean-init

```
$ pixi run cgitsync clean-init <spec.cgs> [--output-path DIR] [--commit-  
  gitignore]  
  [--force-gitignore-sync] [--git-user-name NAME] [--git-user-email  
  EMAIL]  
  [--force-protocol {ssh,https}]
```

Recovery-oriented initialisation for a `.cgs` source. It follows the same high-level pipeline as `initialise(.cgs)` (including the `.gitignore` lifecycle sync and `-force-protocol` behaviour described there), but inserts a cleanup phase before cloning:

load->expand->validate->purge->clone->gitignore

Use it when an earlier initialisation attempt left partial child checkouts. `CGSHOME/.cgitsync/master.toml` (the persisted Git identity override, if any) is preserved by the purge phase — it is workspace-local configuration, not generated clone state.

4.7 purge

```
$ pixi run cgitsync purge <spec.cgs> [--output-path DIR]
```

Removes generated clone state from the resolved `CGSHOME` without cloning. It deletes child repository directories declared directly under the project root and project-local `*.lgr` files. Nested directories that are not immediate root children, and `.cgitsync/master.toml` (see `clean-init` above), are left untouched by this command.

4.8 pull

```
$ pixi run cgitsync pull [spec.cgs|snapshot.gts] [--commit-gitignore]  
  [--force-gitignore-sync] [--git-user-name NAME] [--git-user-email  
  EMAIL]
```

Resynchronises the project tree. `.cgs` sources reload the source, discover nested configs, then resynchronise the loaded tree. `.gts` sources load the saved registry as the starting point and then run the same pull workflow. In both cases the operation starts at the project root: `root -> parent -> leaf`. Every repository — root, parent, and leaf alike — is a plain independent clone and receives its own `git pull -ff-only`. Use `launch-release` when you want to check out a frozen recorded release state instead of pulling current remotes. If local changes block this safe pull, the CLI prints `You can try cgitssync pull-force` command.

For a `.cgs` source, `pull` also runs the same `.gitignore` lifecycle sync described under `initialise` — including `-commit-gitignore`, `-force-gitignore-sync`, and the `-git-user-name/-git-user` identity override — once the tree-wide pull completes. A `.gts` source runs no discovery, so there is nothing new for the sync to find. `pull-force` does not run this sync at all; it is a destructive recovery command, not a lifecycle path the sync is wired into.

4.9 pull-force

```
$ pixi run cgitssync pull-force [spec.cgs|snapshot.gts]
```

Destructively resynchronises the same tree. Every repository — root, parent, and leaf alike — is forced to the selected remote branch, parent-first, with `git fetch`, `git checkout -B <branch> FETCH_HEAD`, and `git clean -fd`. This command can discard local uncommitted changes and untracked files; it never force-pushes.

4.10 clone

```
$ pixi run cgitssync clone <spec.cgs> [--target-dir DIR] [--output-path DIR]
```

Clones the full repository tree declared by the `.cgs` file. This CLI command delegates to `ComplexGitSyncClient.clone`. When `-target-dir` is omitted, the project root is derived from the specification. `-output-path` provides the parent directory used to derive the project root.

For each repository, `cgitssync` tries the repo target branch first and falls back to the repo fallback branch when the preferred branch does not exist on the remote. Nested `.cgs` files are discovered after parent repositories are cloned, so explicit and automatic nested trees are materialised during the same command.

4.11 checkout

```
$ pixi run cgitssync checkout <branch> [--gts <snapshot.gts>] [--ref-kind branch|tag]
```

Checks out a branch or tag across the whole tree, parent-first. Loads the `READY` registry from `-gts`, then propagates the ref, creates missing local branches, and runs `git checkout` in every repo. On success the tree remains `READY` and a new `.gts` snapshot is written.

4.12 add

```
$ pixi run cgitssync add [--gts <snapshot.gts>] [--dry-run]
```

Stages all changes across the tree (`git add -all`), leaf-first. Requires `READY`; tree stays `READY`. With `-dry-run`, `cgitssync` prints the leaf-first execution plan and does not mutate repositories.

4.13 commit

```
$ pixi run cgit sync commit <message> [--gts <snapshot.gts>] [--no-stage]
  [--dry-run]
```

Commits changes across the tree, leaf-first. Loads the READY registry from `-gts`. Repositories with no staged changes after the optional `git add -all` step (suppressed by `-no-stage`) are silently skipped. Requires a READY tree; the tree remains READY on success. Commit preflight now emits warnings for actionable workspace issues such as dirty worktrees, ahead branches, or stale recorded `commit_sha` values. With `-dry-run`, `cgit sync` prints the planned stage/commit sequence without performing git writes.

The commit message conventionally ends with `CGS#VERSION`.

4.14 push

```
$ pixi run cgit sync push [--gts <snapshot.gts>] [--dry-run]
```

Pushes all repositories to their configured remotes, leaf-first. Loads the READY registry from `-gts`. Each repo is pushed to `entry.remote_name` (defaulting to "origin") on its currently resolved branch. Refreshes the stored hash in `GitTree`. Requires a READY tree; the tree remains READY on success. Push preflight blocks detached HEADs, missing remotes, unresolved merges, and behind/diverged branches; it warns when the local branch is ahead or when the loaded snapshot records an older `commit_sha`. With `-dry-run`, `cgit sync` prints the push plan without mutating any repository.

4.15 freeze-release

```
$ pixi run cgit sync freeze-release <name> <message> [--gts <snapshot.gts>]
  [--dry-run]
$ pixi run cgit sync freeze-release-force <name> <message> [--gts <
  snapshot.gts>] [--dry-run]
```

Minimalist release workflow from a READY tree. It chains public operations in order: `add -> commit -> pull -> push -> freeze`. The force variant uses `pull-force` instead of `pull`. The release name becomes the shared tag and freeze snapshot label; the message is used for the release commit.

The `pull/pull-force` step is skipped (not attempted) when the current branch has no upstream configured yet — e.g. a branch just created and checked out this same session, never pushed before. There is nothing to pull in that case; `push` (already upstream-aware, adding `-u` automatically the same way plain `git push -u` would) publishes the branch on its own. On any branch that already has an upstream, the workflow is unchanged.

4.16 freeze

```
$ pixi run cgit sync freeze <name> [--gts <snapshot.gts>] [--dry-run]
```

Performs a release freeze across all repos, leaf-first: stage/commit, create+push the shared release tag, refresh SHA/state, and write a `.gts` snapshot. Loads the READY registry from `-gts`. Requires READY and leaves the tree READY. Freeze preflight checks remote availability, branch alignment, detached HEADs, unresolved merges, and tag absence. Dirty worktrees and stale recorded `commit_sha` values are reported as warnings because freeze will stage, commit, and snapshot them. Freeze snapshots also include a deterministic `freeze_manifest` section that records freeze invariants (immutable snapshot, validated workspace, synchronized tag,

`release-name`, ledger checkpoint) and declares `launch_state` as the canonical restore operation. Their immutable snapshot filename includes the release, for example `gts-000001-release-2026.05.gts`, while the ledger id remains `gts-000001`. With `-dry-run`, `cgitsync` prints the planned add/-commit/tag/push graph for the loaded workspace without mutating it.

4.17 import-submodules

```
$ pixi run cgitsync import-submodules REPO_ROOT [--apply] [--output FILE]
```

Reads `REPO_ROOT/.gitmodules` and reports (or converts) every declared git submodule to a plain `ComplexGitSync` nested clone.

Without `-apply` (the default), the command performs a *dry run*: it prints each submodule's name, path, URL, and branch, and the `.cgs` entry that would be generated, without touching the repository.

With `-apply`:

- Verifies every submodule's working tree is clean (`git status -porcelain` must be empty); rejects dirty trees immediately.
- Runs `git rm -cached <path>` per submodule, dropping the gitlink from the index while leaving the child's working tree and `.git` directory intact (no re-clone, no local history lost).
- Removes the converted stanza from `.gitmodules` (deletes the file entirely when all stanzas are converted) and stages the change.
- Appends `<path>` to the parent's `.gitignore` using the same helper that the `.gitignore` lifecycle sync uses.

The `-output FILE` option (only effective with `-apply`) writes a validated `.cgs` snippet containing the converted submodule entries to `FILE`, suitable for integration into the project's main `.cgs` file.

After applying, review and commit the staged changes manually:

```
$ git -C REPO_ROOT status          # deleted .gitmodules, modified .gitignore
$ git -C REPO_ROOT commit -m "chore: retire git submodules"
```

Note: `ComplexGitSync` stores no credentials and has no private-repo authentication story. If a submodule's upstream is private and the ambient environment lacks access, the subsequent `cgitsync initialise` or `pull` will fail at the clone step. The `import-submodules` step itself (which only touches the local index and working tree) will still succeed.

4.18 verify

```
$ pixi run cgitsync verify [--search-dir DIR] [--repair]
```

Verifies the hash-chained `.cgitsync/lgr` register for tamper-evidence: every entry's `prev` field must match the previous entry's `entry_hash` (`BROKEN_LINK` if not), every entry's own `entry_hash` must match its recomputed canonical hash (`BAD_ENTRY_HASH` if not), `seq` numbers must be gap-free and unique (`SEQ_GAP/SEQ_DUPLICATE`), and the cached `HEAD` pointer must agree with the recomputed true head (`HEAD_STALE`). A register with no entries yet — the common case, since nothing currently writes to this format — is reported clean; that is not itself a finding.

`CGSHOME` is located the same way as `status`: from `-search-dir`, else `$CGSHOME`, else the current working directory's ancestors. Exits 0 when clean, 1 when any finding is reported.

With `-repair`, a stale HEAD cache is corrected to match the recomputed true head. **Entries themselves are never rewritten or deleted** by `verify`, repair or not — a broken chain is reported, not silently healed, since a register that can be edited back to looking clean provides no evidence of anything.

Scope note: this checks chain linkage and the HEAD cache only. Cross-referencing entries against the actual `state(<hash>)_n/` directories on disk (`MISSING_STATE`, `ORPHAN_STATE`, `STATE_DIGEST_MISMATCH`) is not implemented yet.

4.19 launch-release

```
$ pixi run cgitssync launch-release <name> [--gts <snapshot.gts>]
```

Checks out the frozen release tag `<name>` across the loaded READY tree, parent-first, and writes a fresh `.gts` snapshot. This is the release oriented CLI path for returning a workspace to a frozen version; direct tag creation remains available from the Python client API as `tag()`.

The integration suite validates local no-network lifecycle coverage through the CLI path, including initialisation and the READY git command cycle `add -> commit -> push -> freeze -> launch-release`.

4.20 view-tree

```
$ pixi run cgitssync view-tree [spec.cgs|snapshot.gts] [--depth N] [--collapse REPO]
```

Renders a deterministic terminal tree that focuses on repository topology and operational state: `name` `[branch]` `local_state` `sync_state`. Use `-depth` to limit visible levels and repeat `-collapse` to hide selected subtrees.

Branch Topology Propagation Rules (T35):

1. **Reference branch:** The root repository's current branch is the canonical reference for all repos.
2. **Leaf-to-root inheritance:** Branch targeting flows root-first via `propagate_global_branch`, verified on-disk by `ComplexGitSyncClient.validate_topology()` (Python API only) without issuing any git writes.
3. **Allowed divergence:** Repositories in a frozen (TAG) state produce a `tag_divergence` diagnostic that does *not* make the topology incoherent.
4. **Incoherent states** (blocking): `misaligned_branch` (different branch from root) and `detached_head` (detached HEAD without a tag reference).

4.21 status

```
$ pixi run cgitssync status [--gts FILE]
```

Summarises tree readiness and per-repository sync state. The table reports the local branch, upstream branch, local worktree state, ahead/behind tracking counts, current HEAD, and the commit recorded in the loaded `.gts`.

5 Document Formats

`.cgs` means **ComplexGitSync** specification, `.gts` means **GitTreeState** snapshot, and `.lgr` means **Local Git Register**.

5.1 Local Authoring Spec (.cgs)

A TOML file that describes the repository tree to manage. It is the only hand-authored input; it is never modified at runtime. The normal form has two top-level keys:

```
project = "CGS11"
repos = [
  "gitlab:CGS_test/CGS11",
  "gitlab:CGS_test/CGS12",
  "github:flipoyo/CGS1h1",
]
```

Repository identifiers have the form `provider:owner/repository`. Nested GitLab namespaces are supported; the final path segment is the repository name. The document pipeline is:

PARSE -> NORMALIZE -> VALIDATE -> canonical CgsDocument

The canonical providers and hosts are:

Provider	Host	Remote generation
github	github.com	SSH and HTTPS
gitlab	gitlab.com	SSH and HTTPS
codeberg	codeberg.org	SSH and HTTPS
custom	explicit URL required	SSH and HTTPS

Repository-ID parsing is provider-independent and deterministic. For example:

```
codeberg:GX4G/GX4G
```

normalizes to `gitprovider = codeberg`, `project_owner_name = GX4G`, and `repo_name = GX4G`. A custom identifier must use an inline table so the host is explicit:

```
{ repository = "custom:team/tool", gitprovider_url = "https://git.
  example.com" }
```

No host is inferred for `custom`. The configured access protocol then selects `git@host:owner/repository.git` or `https://host/owner/repository.git`.

The textual grammar and its only parser, `parse_repo_id()`, live in `cgs_format.py`. The provider model in `git_repo.py` receives the already-separated provider, owner, and repository fields and composes remotes; it does not parse authoring shorthand.

The `.cgs` format pipeline is deterministic and offline. A syntactically valid repository may therefore reference a host, owner, branch, or tag that is not currently reachable. Remote existence and reference availability are checked only by explicit Git operations during runtime resolution and execution.

Normalization supplies `default_branch = main`, makes `fallback_branch` equal to that default, selects `access_protocol = ssh`, enables `nested_config = auto`, and uses the repository name as `relative_path`. If exactly one repository name matches `project`, its path is inferred as ..

Advanced inline tables are used only where a value differs from those defaults:

```
project = "demo"
repos = [
  "github:owner/demo",
  { repository = "github:owner/docs", relative_path = "documentation",
    access_protocol = "https" },
]
```

The legacy advanced `[project]` and `[[repos]]` tables remain accepted for project-wide branch settings, custom provider URLs, tags, and other explicit overrides. Unknown providers, malformed identifiers, duplicates, and missing `project` or `repos` are rejected before a `GitTree` is built.

Two checked-in files document this boundary. Copy `examples/template.cgs` for normal authoring. Its equivalent `examples/normalized_template.cgs` expands every inferred field and is provided as a developer reference for the canonical `CgsDocument` shape. The two files are regression-tested for semantic equivalence.

The repository also keeps three checked-in, didactically ordered tutorials under `docs/tutorials/` — easiest first, see `docs/tutorials/README.md` for the guided order — exercising this authoring surface end to end: `docs/tutorials/01_first_multi_repo_workspace.md` walks a fully exercised local sandbox topology (`CGS111`) through the complete CLI lifecycle; `docs/tutorials/02_onboarding_a_hand-authors a .cgs` for the larger, real-world `cawaqs` tree and the point where `cgitsync` hands off to the project's existing `make`-based build workflow; and `docs/tutorials/03_configuration_discovery_modes.md` covers the real-world `cawaqsviz` project (not synthetic) through all three ways `ComplexGitSync` can arrive at a working `.cgs` for a project it does not already control the source of.

`examples/cawaqsviz.cgs`, one of that third tutorial's routes, describes `https://gitlab.com/cawaqs/gviz/cawaqsviz` and exercises several patterns above at once: a nested GitLab subgroup path (`cawaqs/gviz/cawaqsviz`, three segments rather than a plain `owner/repository` pair), an explicit `relative_path = "."` on the root entry rather than relying on the project-name auto-mount rule, two GitHub children mounted at non-default relative paths (`external/HydrologicalTw` and `docs/CWV_user_guide`). Neither child has a `.cgs` of its own, but no override is needed for that: the default auto nested discovery resolves a repository with zero `*.cgs` matches as a normal leaf. This file previously shipped with a nonexistent repository identifier and an invalid `default_branch`; it was corrected and given real end-to-end test coverage (`tests/integration/test_cgsi_top` and one verified run against the live repositories) as `AgentSpec/Onboarding_DevPlanTicket.md` Phase 1. See `docs/tutorials/03_configuration_discovery_modes.md` for the full worked example, and `bootstrap` above for running `ComplexGitSync` against it without cloning `ComplexGitSync` inside the tree it manages.

Generated specifications use the same concise representation. The `GitTree.to_cgs()` convenience method delegates model projection to `cgs_format.py`; neither `GitTree` nor `GitRepo` formats TOML or implements the authoring grammar. Serialization omits every default that normalization can deterministically reconstruct and retains inline tables only for exceptional settings such as a non-default branch, path, protocol, tag, custom host, or discovery rule.

The supported round trip is semantic:

```
.cgs -> CgsDocument -> GitTree -> CgsDocument -> .cgs
```

Parsing and normalizing the generated file must reproduce the initial canonical document. Whitespace, comments, and TOML table layout need not be byte-for-byte identical.

5.2 Repository Placement and Nested Discovery

For every non-root entry, `relative_path` is relative to the parent repository represented by the current `.cgs`; it is not relative to the process working directory. At the top level that parent is the project root. While expanding a nested config, it is the repository containing that config. The default is the repository name. A unique repository matching the project name uses `.` because it represents the current root.

The `nested_config` option controls whether traversal continues after the repository is available:

- `auto` (default) loads the sole root-level `*.cgs` file, if one exists; finding none resolves the repository as a normal leaf; multiple matches are rejected as ambiguous.

- `disabled` does not inspect the repository for nested config.
- A relative path ending in `.cgs` loads that exact file inside the repository, failing if it does not exist. It may not escape the repository root.

For the integration topology, `CGSil2.cgs` contains:

```
{ repository = "github:flipoyo/CGSih1", relative_path = "../CGSih1" }
```

Because the config describes the tree below `CGSil2`, the path resolves from that directory:

```
<CGSHOME>/CGSil2/../CGSih1 -> <CGSHOME>/CGSih1
```

It therefore references the existing sibling declared by `CGSil1.cgs`, instead of cloning another `CGSih1` under `CGSil2`. Discovery recognizes the already-registered absolute path and keeps the canonical entry — this guard runs before `auto` would ever get a chance to reopen `CGSih1.cgs` through the cross-reference, so no `nested_config` override is needed here.

5.3 Git Tree State Snapshot (`.gts`)

A TOML file generated automatically by bootstrapping and release operations. It must never be hand-edited. It captures the exact commit SHAs and lifecycle state of the tree at a point in time.

Top-level sections:

- `[document]` — `format_version`, `schema_version`, `generated_at`, `command_origin`, `hash_algorithm`, `snapshot_hash`.
- `[project]` — project name and `root_absolute_path`.
- `[tree_state]` — `lifecycle_state`, `is_ready`, `registry_complete`.
- `[[repo_state]]` — one entry per repo; includes `commit_sha`, `sync_state`, and ref information.

Why both version fields and hash metadata exist:

- `format_version` identifies the broad document family and keeps long-range compatibility intent explicit.
- `schema_version` identifies the exact field-level contract used by validation and canonical snapshot serialization.
- `snapshot_hash` (with `hash_algorithm = "sha256"`) provides deterministic workspace identity: identical tree state produces identical digest even if volatile metadata (`generated_at`, `command_origin`) differs.

How this is used at runtime:

- During snapshot generation, `ComplexGitSync` writes `schema_version`, `hash_algorithm`, and canonical `snapshot_hash`.
- During load/validation, `snapshot_hash` is recomputed from the canonical payload and must match.
- The project-local `.lgr` register uses this canonical hash to deduplicate equivalent snapshots.

5.4 Local Git Register and Sync Ledger (.lgr)

A TOML file maintained per project at `<project-root>/<Project_name>.lgr`. It is updated automatically whenever a `.gts` snapshot is written.

Sections:

- `[register]` — current snapshot pointer (`current_snapshot_id`, `current_snapshot_hash`, `current_snapshot_path`).
- `[[snapshots]]` — list of stable `gts-XXXXXX` identifiers, one per unique workspace state, deduplicated by SHA-256 hash.
- `[[ledger]]` — append-only list of synchronisation events forming a DAG via `parent_sync_ids`.

Ledger event schema:

```
[[ledger]]
sync_id      = "lgr-000001"
parent_sync_ids = []
operation    = "clone"
timestamp    = "2026-05-20T19:48:50.159Z"
actor        = "username"
workspace_hash = "5f7c8a2d..." % 64-character SHA-256 hex digest
gts_snapshot_id = "gts-000001"
affected_repos = ["demo", "dep-a"]
```

The `workspace_hash` field equals the canonical SHA-256 digest (`document.snapshot_hash`) of the associated `.gts` file, creating a direct link between each ledger event and the immutable workspace state it records. Events are parents-before-children in DAG order.

6 Worked Examples: From CGSill to cawaqsviz

ComplexGitSync ships four checked-in `examples/` topologies, and a same-numbered tutorial under `docs/tutorials/` for each level below (`docs/tutorials/README.md` lists all three in order). Run through them in order: each one adds exactly one new idea on top of the last, so together they form a guided path from the simplest possible `.cgs` to a real project with no `.cgs` of its own at all.

Every command below is a Pixi task. `cgitsync` is not a globally installed executable — always run `pixi run cgitsync ...`, never a bare `cgitsync ...` (see the §1 “CLI Overview” note above).

Level	Topology	Entry point	New idea introduced
1	CGSill	hand-authored <code>.cgs</code>	minimal spec, name-matching auto-mount
2	cawaqs	hand-authored <code>.cgs</code>	the same minimal-field style as Level 1, at 19-repo scale
3a	cawaqsviz	hand-authored <code>.cgs</code>	explicit overrides on a real, irregular tree
3b	cawaqsviz	discover	drafting a <code>.cgs</code> from a check-out, no authoring at all
3c	cawaqsviz	import-submodules	migrating an existing git-submodules project

Levels 3a–3c are not three different projects: they are three different *starting points* for the same real project, `cawaqsviz`, chosen so this section can demonstrate every route ComplexGitSync

offers for arriving at a working `.cgs` — write it by hand, derive it from a checkout, or migrate it from an existing submodules setup — without inventing a fourth toy topology to do it. It is the most advanced level not because `cawaqsviz` is a larger tree than `cawaqs` (it is much smaller, three repos against nineteen) but because, unlike either earlier level, it has no `.cgs` anywhere upstream to copy from.

6.1 Level 1 — CGSil1: the minimal synthetic topology

CGSil1 is the easiest way into ComplexGitSync: a 4-repository, mixed-provider tree with every field left at its default. Its full `.cgs` and the rationale for each line are in §1 “Local Authoring Spec” above; `docs/tutorials/01_first_multi_repo_workspace.md` carries the full transcript with expected output for every step. What follows is the command sequence itself.

```
# 1. Fetch the root repo and cgitSync itself (nested mode: cgitSync is
#     cloned inside the tree it will manage).
$ git clone https://gitlab.com/CGS_test/CGSil1
$ cd CGSil1 && git clone https://github.com/flipoyo/ComplexGitSync
$ cd ComplexGitSync

# 2. Check the spec offline, no network beyond parsing.
$ pixi run cgitSync validate ../CGSil1.cgs

# 3. Preview the topology before touching disk.
$ pixi run cgitSync view-tree ../CGSil1.cgs

# 4. Clone every declared repo and reach READY.
$ pixi run cgitSync initialise ../CGSil1.cgs

# 5. The ordinary git cycle, run tree-wide from here on --- no path
#     argument needed, the .gts snapshot is discovered automatically.
$ pixi run cgitSync pull
$ pixi run cgitSync add
$ pixi run cgitSync commit "my commit message"
$ pixi run cgitSync push

# 6. Minimalist release: stage, commit, pull, push, freeze in one call.
$ pixi run cgitSync freeze-release v1.1.0 "release v1.1.0"

# 7. Return to a frozen release at any later point.
$ pixi run cgitSync launch-release v1.1.0
```

The only `.cgs` idea this level relies on is the **name-matching auto-mount**: because CGSil1’s repository identifier’s last segment equals `project = "CGSil1"`, the root is inferred at `relative_path = "."` with no override written anywhere. That convenience is exactly what stops being safe once a project’s identifier and display name diverge — which is where Level 3a picks up.

6.2 Level 2 — cawaqs: the same minimal style, at scale

`cawaqs` scales Level 1’s minimal, no-override authoring style up to a real 19-repository build tree (root + 17 physics libraries across five GitLab groups + one required build-tooling repo), instead of a synthetic 4-repo one. Every entry’s on-disk layout already matches the default `relative_path = <repo_name>`, so not a single `relative_path` override is written anywhere in `examples/cawaqs.cgs` — only the shared branch-fallback convention. Full narrative in `docs/tutorials/02_onboarding_a_real_build_tree.md`.

```
$ pixi run cgitSync validate examples/cawaqs.cgs
```

```

$ pixi run cgitssync bootstrap examples/cawaqs.cgs cawaqs-smoke-test
$ pixi run cgitssync view-tree --search-dir <path bootstrap printed>

# Move the whole 19-repo tree to a shared branch in one call, falling
# back to main per-repo wherever that branch does not exist.
$ pixi run cgitssync checkout my-feature-branch

```

Reaching **READY** here only replaces the *repository-fetching and branch-selection* half of **cawaqs**’s own `make_Cawaqs.sh/make_Cawaqs_from_branches.sh` scripts. **ComplexGitSync** does not compile anything: the remaining `make -f Makefile` build step is still the user’s job, and only succeeds from this tree with `LIB_HYDROSYSTEM_PATH` left unset or pointed at the bootstrapped root — the alternative shared, decoupled install layout those scripts also support is out of scope for **ComplexGitSync**’s containment model.

discover (Level 3b below) is not a route into **cawaqs.cgs**: the 17 library repositories never coexist inside one directory tree until *after* a **.cgs** already lists them, so there is no single checkout for a filesystem walk to scan. Authoring this one from documented developer knowledge, the way Level 3a does for **cawaqsviz**, is the only available route.

6.3 Level 3 — **cawaqsviz**: three ways to the same **.cgs**

cawaqsviz (<https://gitlab.com/cawaqs/gviz/cawaqsviz>) is a real GitLab project, nested three path segments deep, with two GitHub children that were, until recently, plain git submodules. It has no **.cgs** of its own anywhere upstream. That makes it a good vehicle for showing all three ways **ComplexGitSync** can produce a working **.cgs** for a project it does not already control the source of, ordered here by how much the operator has to already know about the tree before running the command. Full narrative in `docs/tutorials/03_configuration_discovery_modes.md`.

6.3.1 3a — Write it by hand

The most explicit, and most error-prone, route: read the project’s own **.gitmodules** and topology, then author `examples/cawaqsviz.cgs` directly. Its corrected content and the exact mistakes the first draft made (wrong nested-subgroup identifier, missing `relative_path` overrides) are already spelled out in §1 “Local Authoring Spec”; this level is about running it, not re-deriving it:

```

$ pixi run cgitssync validate examples/cawaqsviz.cgs
$ pixi run cgitssync bootstrap examples/cawaqsviz.cgs cawaqsviz-demo
$ pixi run cgitssync view-tree --search-dir <path bootstrap printed>

```

bootstrap is used instead of **initialise** here because, unlike **CGSill** above, this example is not meant to have **ComplexGitSync** cloned inside the tree it manages — see **bootstrap** in §1 for why the two commands pick different default locations.

6.3.2 3b — Derive it with **discover** (autodiscover mode)

The opposite route: start from nothing but a checkout, and let **ComplexGitSync** read the filesystem instead of writing the **.cgs** by hand. This is the route to prefer whenever a checkout is already available, since it cannot repeat the hand-authoring mistakes of 3a — `relative_path` falls out of the walk directly instead of being typed.

```

# A plain clone leaves submodule paths empty; initialise them first so
# discover has something to find at those paths.
$ git clone https://gitlab.com/cawaqs/gviz/cawaqsviz cawaqsviz-scan
$ cd cawaqsviz-scan && git submodule update --init --recursive

# Dry run: report what discover sees, write nothing.

```

```
$ pixi run cgitSync discover . --max-depth 3

# Satisfied with the report: draft the .cgs.
$ pixi run cgitSync discover . --write cawaqsviz-discovered.cgs
$ pixi run cgitSync validate cawaqsviz-discovered.cgs
```

The draft reconstructs 3a’s hand-corrected file field-for-field: the root at `relative_path = "."` and both children at their real submodule paths (`external/HydrologicalTwinAlphaSeries`, `docs/CWV_user_guide`) rather than their bare repo names. Neither child has a `.cgs` of its own, but `discover` writes no `nested_config` override for that: the default "auto" already resolves a repository with zero nested `*.cgs` matches as a normal leaf. See `discover` in §1 for exactly what is and is not reported (no network calls, unresolvable remotes are warned about and left out, never guessed at).

6.3.3 3c — Migrate it with import-submodules

The historical route: before either `.cgs` above existed, `cawaqsviz` tracked both children as real git submodules. This mode starts from that state and converts it, rather than describing a tree that is assumed to already be plain clones.

```
# Dry run against a real cawaqsviz checkout that still has submodules:
# reports name, path, URL, branch per submodule; changes nothing.
$ pixi run cgitSync import-submodules /path/to/cawaqsviz

# Convert: drop the gitlinks, retire .gitmodules, extend .gitignore,
# and emit the equivalent .cgs entries.
$ pixi run cgitSync import-submodules /path/to/cawaqsviz \
  --apply --output cawaqsviz_submodules.cgs

# Review and commit the resulting working-tree changes as usual.
$ cd /path/to/cawaqsviz && git status
$ git commit -m "chore: retire git submodules in favour of
  ComplexGitSync"
```

The full before/after picture (`index`, `.gitmodules`, `.gitignore`) and the automated test that proves it against fixture repositories are in `docs/tutorials/03_configuration_discovery_modes.md` §4 — this level is the condensed command sequence, not a restatement of that walkthrough. `import-submodules` and `discover` answer different questions about the same tree: `discover` reports what is *checked out*, `import-submodules` reads what git *declares* in `.gitmodules` and acts on it — a submodule path that was never initialised is invisible to 3b but not to 3c.

6.3.4 A shared caveat across all three

`ComplexGitSync` stores no credentials and has no private-repository authentication story (§1 “import-submodules” and “discover”). All three routes above assume every repository in the tree is reachable anonymously or through credentials the ambient environment (`ssh-agent`, an HTTPS credential helper) already holds.

7 Configuration and Environment

`ComplexGitSync` uses project files for topology and does not require global topology configuration. Snapshot discovery can use `CGSHOME`; when it is not set, the `initialise(.cgs)` output-path default is `CGSPATH=../..` (nested mode: `ComplexGitSync` cloned inside the tree it manages); later commands can also walk upward from the current directory to find `.cgitsync/state/bootstrap(.cgs)` uses a different default instead — `CGSPATH=$HOME/.cgs/CGS<timestamp>/`,

created automatically — since it is meant for a `ComplexGitSync` clone that is not nested inside any project tree; see `bootstrap` above. The logging and runtime-state subsystems use the user state directory by default:

- logs: `/.local/state/ComplexGitSync/logs/`
- latest runtime-state mapping: `/.local/state/ComplexGitSync/runtime/`

If `XDG_STATE_HOME` is set, these directories are rooted there instead.

8 Error Reference

ConfigValidationError Raised when `project` or `repos` is missing, a repository identifier is malformed, or a canonical field is invalid (for example an unknown `gitprovider`).

GitSyncError Raised when clone-time Git operations fail, for example because the remote branch cannot be resolved or the target directory is already occupied.

Exit code 1 General CLI failure (parse error, validation error).

Exit code 2 Command exists but is not yet implemented.

9 Troubleshooting

“gitprovider_url is required for custom provider” Add a `gitprovider_url` field to the offending `[[repos]]` entry in your `.cgs` file.

validate exits non-zero despite a correct-looking file Check that `project` is present and every `repos` entry uses `provider:owner/repository`. For advanced `[[repos]]` blocks without a `repository` shorthand, include `project_owner_name` and `project_name`.

Clone destination already exists and is not empty Re-run clone with `-target-dir` to force a clean location, or let `cgitsync` choose a suffixed default directory.

No cloneable branch found for <repo> Confirm that the remote exposes either the repo’s `default_branch` or its `fallback_branch`.

PyYAML import error when calling from_yaml/to_yaml Add PyYAML in the Pixi environment: `pixi add pyyaml`.